

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 9: Dead Reckoning & Flight Computer Principles (CRP-5)

Syllabus Source: CAE Oxford (pp. 325-364) | Pass Target: $\geq 90\%$ AviationExam

1. DEAD RECKONING (DR) FUNDAMENTALS

Dead Reckoning is the navigation method of calculating current and future aircraft position from a previously known position using heading, airspeed, wind, and elapsed time:

- The Navigation Vector Triangle: Heading Vector (True Heading + TAS) + Wind Vector (Wind Direction + Wind Speed) = Track Vector (True Track + Groundspeed).
- The CRP-5 / E6B Slide Rule: Circular slide rule solving: 1. Time-Speed-Distance (Time = Dist / GS); 2. Fuel consumption (Burn = Time x Fuel Flow); 3. TAS from Calibrated Airspeed (CAS), Pressure Altitude, and Temperature (correcting for density error and compressibility).

2. TRUE AIRSPEED (TAS) VS INDICATED AIRSPEED (IAS)

- Rule of Thumb: TAS INCREASES by approximately 1.5% to 2.0% per 1,000 ft of altitude above sea level for a given Indicated Airspeed!
- Mach Number & Temperature: Mach = TAS / Speed of Sound (a). Speed of Sound depends SOLELY on Absolute Temperature (Kelvin): $a = 38.94 \times \sqrt{T_K}$ knots. In colder air, sound travels slower!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Estimating TAS from IAS and Altitude

SCENARIO (Basic Navigation Rule of Thumb Drill): An aircraft climbs to FL 100 (10,000 ft): - Indicated Airspeed (IAS / CAS) = 150 knots - Outside Air Temperature is standard ISA (+15°C - 20°C = -5°C) QUESTION: What is the approximate True Airspeed (TAS)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Recall the standard rule of thumb for TAS increase with altitude: - Rule: Add 2% to IAS per 1,000 ft of altitude! Step 2: Calculate the total percentage increase for 10,000 ft: - Altitude = 10,000 ft = 10 x 1,000 ft. - Total percentage increase = 10 x 2% = +20%! Step 3: Calculate the knots to add to IAS: - 20% of 150 knots = 0.20 x 150 kt = 30 knots. Step 4: Add the increase to the IAS: - TAS = IAS + Increase = 150 kt + 30 kt = 180 knots! FINAL ANSWER: True Airspeed (TAS) is approximately 180 knots (a 30 knot gain over indicated speed due to lower air density at 10,000 ft).

[!] EXAM TRAP (AVIATIONEXAM) - The Speed of Sound Depends Only on Temperature Trap

- The Speed of Sound DOES NOT depend on pressure or air density! - It depends SOLELY on absolute temperature: $a = 38.94 \times \sqrt{T_{\text{Kelvin}}}$! - Exam questions asking if sound speed changes with pressure are traps: the answer is NO, only temperature matters!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the primary physical reason why True Airspeed (TAS) is greater than Indicated Airspeed (IAS) at high altitude?

[A] Engine develops more thrust [B] Air density DECREASES with altitude, requiring the aircraft to travel faster through the thin air to produce the same dynamic pressure ($q = 0.5 \rho V^2$) [C] Less gravity [D] Instrument error

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The ASI measures dynamic impact pressure. At lower density, the aircraft must fly faster to register the same indicated pressure.

Q2: On which single atmospheric parameter does the local SPEED OF SOUND depend?

[A] Atmospheric pressure [B] ABSOLUTE TEMPERATURE (in Kelvin) [C] Air density only [D] Wind speed

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. In a perfect gas, sound speed is given by $a = \sqrt{\gamma \times R \times T}$. Only absolute temperature T affects sound velocity.

Q3: If an aircraft is cruising at FL 300 with an IAS of 240 kt in standard ISA conditions (-45°C), what is the approx TAS?

[A] 300 kt [B] Approx 384 kt ($240 \text{ kt} + 60\% = 384 \text{ kt}$) [C] 240 kt [D] 450 kt

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. $30,000 \text{ ft} \times 2\% / 1,000 \text{ ft} = +60\%$. $240 \times 1.60 = 384 \text{ knots TAS}$.